

6V DC Regulated Power Supply

using a 12V AC Step-Down Transformer & 7806 Voltage Regulator IC

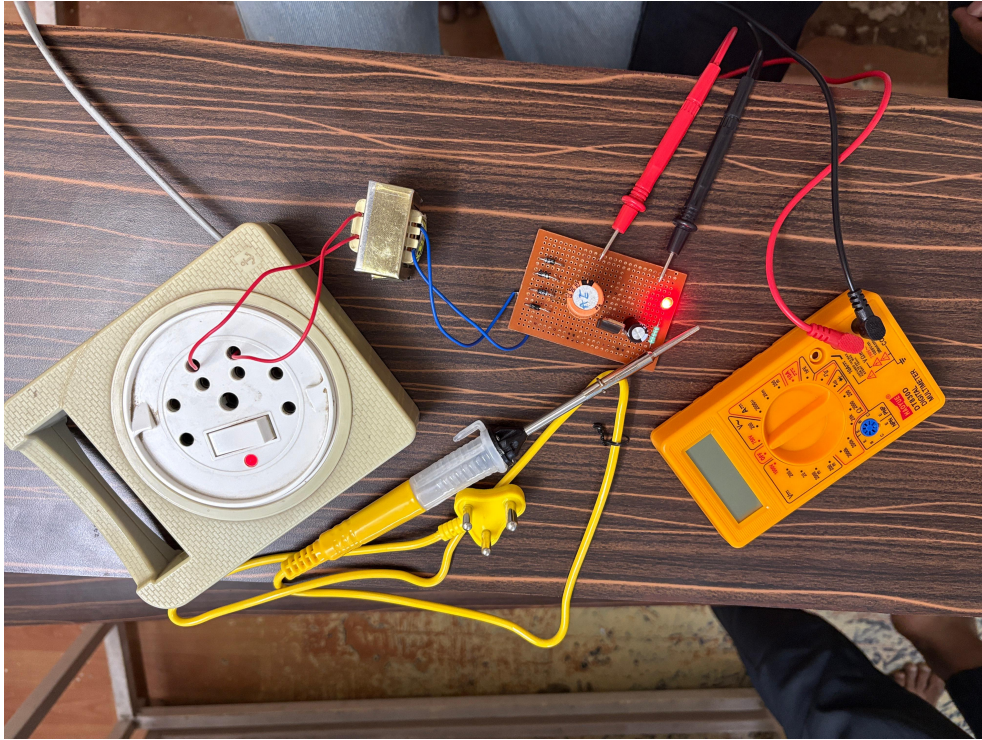


Fig 1: Assembled circuit under test with a digital multimeter

Project Report

1. Introduction

This project is a regulated DC power supply that converts 230V household AC mains into a steady, clean 6V DC output. It uses a step-down transformer to bring the mains voltage down to a safe 12V AC level, a bridge rectifier to convert that AC into pulsating DC, a filter capacitor to smooth the ripples, and a 7806 fixed-voltage regulator IC to hold the final output at a constant 6V DC regardless of small variations in input voltage or load current. An LED indicator confirms that the supply is powered and working, and the output is verified using a digital multimeter.

1.1 Objective

To design and test a simple, reliable linear power supply circuit that produces a fixed 6V DC output suitable for powering low-voltage electronic circuits, microcontroller boards, or sensor modules, starting directly from AC mains.

This circuit works directly from AC mains voltage (230V). Mains-connected sections must be properly insulated and handled only with correct safety precautions; this report is for educational documentation purposes.

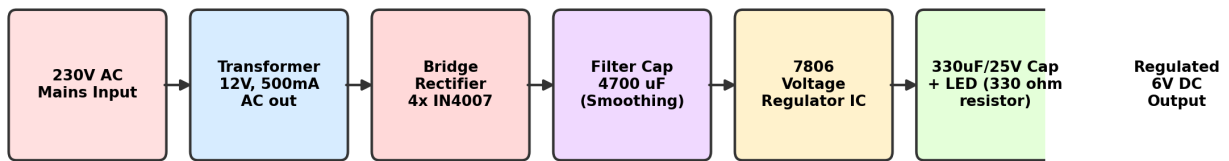
2. Components Used

S.No	Component	Quantity	Purpose
1	Step-down Transformer (230V AC -> 12V AC, 500 mA)	1	Reduces mains voltage to a safe AC level
2	IN4007 Rectifier Diode	4	Bridge rectifier - converts AC to pulsating DC
3	Filter Capacitor - 4700 uF (Electrolytic)	1	Smooths rectified DC ripples before the regulator
4	7806 Voltage Regulator IC	1	Regulates output to a fixed 6V DC
5	Output Capacitor - 330 uF, 25V	1	Further smooths the regulated 6V output
6	LED	1	Power-on indicator
7	Resistor - 330 ohm	1	Current-limiting resistor for the LED
8	Perforated (Vero) Board	1	Circuit assembly
9	AC Mains Plug & Wires	1 set	Connects circuit to 230V supply
10	Digital Multimeter	1	Measuring and verifying output voltage

3. Circuit Diagram & Stages

The power supply is built in six stages, each converting or conditioning the power a step further until a clean, regulated 6V DC is available at the output terminals.

6V DC Regulated Power Supply - Block Diagram

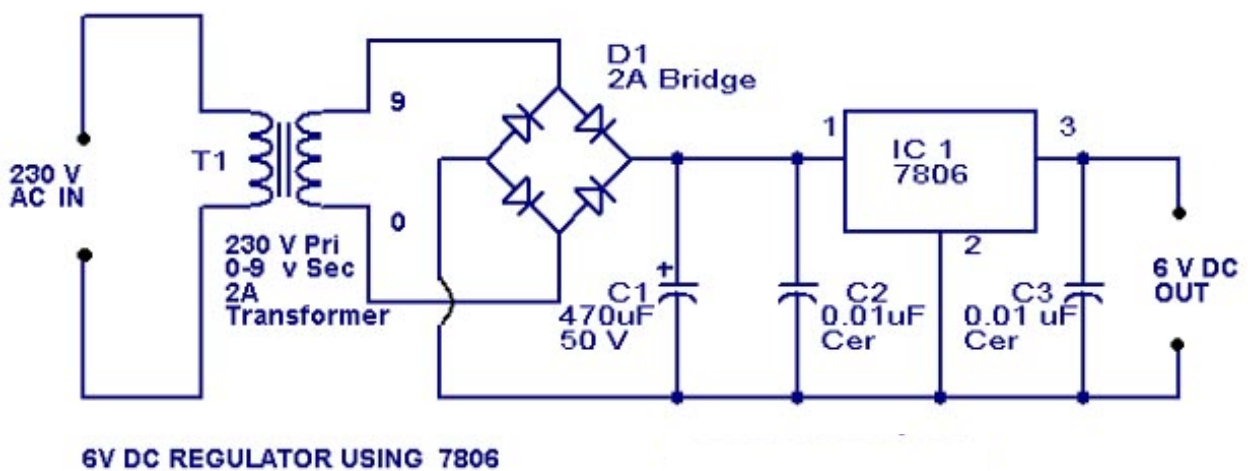


Testing: Multimeter connected across the 7806 output measures the regulated 6V DC delivered to the load.

Fig 2: Block diagram of the power supply stages

3.1 Stage-wise Description

Stage	Function
230V AC Mains Input	Raw household AC supply feeding the transformer primary
Step-down Transformer (12V, 500 mA)	Steps 230V AC down to a lower 12V AC on the secondary winding
Bridge Rectifier (4x IN4007 diodes)	Converts the 12V AC into pulsating (unregulated) DC
Filter Capacitor (4700 uF)	Smooths the pulsating DC into a less-rippled DC waveform
7806 Regulator IC	Regulates the smoothed DC to a fixed, stable 6V DC output
Output Capacitor (330 uF, 25V)	Further filters output ripple
LED (3mm red)	LED shows power is present
6V DC Output	Final regulated output available to power the load circuit



4. Working Principle

The 230V AC mains supply is first connected to the primary winding of the 12V, 500 mA step-down transformer, which reduces it to a safer 12V AC on the secondary side. This 12V AC is fed into a bridge rectifier made of four IN4007 diodes, which conduct in pairs during each half-cycle of the AC waveform to produce a pulsating DC output at its terminals (both half-cycles of the AC are converted to DC, giving full-wave rectification).

This pulsating DC still contains significant ripple, so a 4700 uF filter (smoothing) electrolytic capacitor is connected across the rectifier output. It charges up near the peak voltage and discharges slowly into the load between peaks, reducing the ripple and producing a much steadier DC level before it reaches the regulator.

This filtered but still somewhat unregulated DC is then fed into the input pin of the 7806 fixed voltage regulator IC. The 7806 internally uses a reference voltage and feedback circuitry to hold its output pin at a constant 6V DC, automatically compensating for small changes in input voltage or the current drawn by the load, as long as the input stays above the minimum voltage the regulator needs to operate correctly (its dropout voltage). A 330 uF, 25V output capacitor further smooths the regulated voltage, and the LED, wired in series with a 330 ohm current-limiting resistor, lights up to visually confirm the supply is live. The final regulated 6V DC is available at the output terminals to power the intended load circuit.

5. Testing & Results

The completed circuit was tested using a digital multimeter set to the DC voltage range, with the probes connected across the 7806's output and ground. The transformer was plugged into a 230V AC mains socket, and the LED indicator lit up, confirming the circuit was powered.

The multimeter reading at the output terminals showed a steady DC voltage close to the rated 6V, confirming that the transformer, bridge rectifier, filter capacitor, and 7806 regulator stage were all functioning correctly together to deliver a stable regulated supply.

6. Precautions

- Ensure all mains-side (230V AC) connections are properly insulated and not touched while powered.
- Observe correct polarity of the electrolytic capacitors; reversed polarity can cause them to fail.
- Attach a small heat sink to the 7806 IC if the load current is significant, since it can get warm during regulation.
- Always verify output voltage with a multimeter before connecting any sensitive load circuit.
- Switch off and unplug the mains supply before making or changing any connections on the board.

7. Conclusion

This project successfully demonstrates a simple, reliable linear regulated power supply that converts 230V AC mains into a stable 6V DC output using a step-down transformer, bridge rectifier, filter capacitor, and a 7806 voltage regulator IC. The multimeter test confirmed the circuit delivers a steady, well-regulated voltage suitable for powering low-voltage electronic circuits and modules. The project

illustrates the fundamental building blocks of linear power supply design: step-down transformation, rectification, filtering, and voltage regulation.

The design can be further improved by adding overcurrent/short-circuit protection (fuse or a current-limiting circuit), a heat sink for the regulator IC under higher loads, an output voltmeter display, or replacing the linear regulator with a switching regulator for higher efficiency. Overall, this project offers a solid practical introduction to analog power electronics and regulated power supply design.